

Given: Due:	Further Maths (Statistics) HW 09	Name: MARK SCHEME
Chapter 2: Poisson Distribution		
1) Accidents occur randomly at a road junction at a rate of 18 every year. The random variable X represents the number of accidents at this road junction in the next 6 months. (a) Write down the distribution of X. (2) (b) Find $P(X > 7)$. (2) (c) Show that the probability of at least one accident in a randomly selected month is 0.777 (correct to 3 decimal places). (3)		

Question (a) and question (b) were well answered although in question (a) a few students wrote down Poisson but forgot to give the value of λ .

Since question (c) is a 'show that' question students were required to write down all the steps needed to reach 0.777. Many students went straight from $1 - P(Y = 0)$ to 0.7769. To gain full marks they were required to write down the figures between these two stages ($1 - 0.2231$).

Question	Scheme	Marks
(a) $X \sim \text{Po}(9)$		M1A1 (2)
(b) $P(X > 7) = 1 - P(X \leq 7)$ $= [1 - 0.3239] = 0.6761$		M1 A1 (2)
(c) [Y = no. of accidents in a month] $Y \sim \text{Po}(1.5)$ $P(Y \geq 1) = 1 - P(Y = 0)$ $= [1 - 0.2231] = 0.7769 (= 0.777 \text{ (3dp)})^*$		B1 M1 A1cso (3)
(d) [A = no. of months with at least one accident] $A \sim B(6, 0.777)$ $P(A = 4) = \binom{6}{4} (0.777)^4 (0.223)^2$ $= 0.2719... \quad \text{awrt } 0.272$		M1 M1 A1 (3)
Notes		(10)
(a)	M1 for Poisson (accept Po). Condone $P(9)$ A1 for mean of 9	
(b)	M1 for writing $1 - P(X \leq 7)$. This may be implied by $1 - 0.3239$ or a correct answer A1 for awrt 0.676	
(c)	B1 $\text{Po}(1.5)$ written or used M1 writing or using $1 - P(Y = 0)$ or $1 - P(Y \leq 0)$ or $1 - e^{-\lambda}$ [may not be Y] A1 for at least $(1 - 0.223)$ or better. No need for final comment.* answer given so 0.777 does not imply all three marks	
(d)	1 st M1 for identifying binomial with $n = 6$ and $p = 0.777$ or better. Condone use of $p = 0.223$. May be implied by $(p)^4(1-p)^2$ $p = \text{awrt } 0.777$ or $\text{awrt } 0.223$ 2 nd M1 Must have ${}^6C_4 (0.777)^4 (1 - 0.777)^2$ A1 for awrt 0.272	



2) A robot is programmed to build cars on a production line. The robot breaks down at random at a rate of once every 20 hours.

(a) Find the probability that it will work continuously for 5 hours without a breakdown.

(3)

Find the probability that, in an 8 hour period,

(b) the robot will break down at least once,

(2)

(c) there are exactly 2 breakdowns.

(1)

(d) A new client wants to see the robot for themselves to be convinced of its reliability. What is the maximum duration the site manager can let the client see the robot in action and be at least 90% confident of a fault-free performance?

(3)

In a particular 8 hour period, the robot broke down twice.

(e) Write down the probability that the robot will break down in the following 8 hour period. Give a reason for your answer.

(2)

Although there were a minority of candidates who were unable to identify the correct distribution to use the majority of candidates achieved full marks to parts (a) (b) and (c). Part (e) seemed to cause substantial difficulty. In part (a) the majority of candidates identified that a Poisson (rather than the Binomial) distribution was appropriate but some calculated the parameter as 2.5 or 4 rather than 0.4. A few used $Po(1)$ and calculated $P(Y=5)$.

In part (b) and part (c) the most common error was to use $Po(2.5)$. The majority of candidates were able to work out $P(X > 1)$ and $P(X = 2)$ using the correct Poisson formula. Many thought that their answer to part (c) was the correct solution while others used or multiplied their answers to both (b) and (c). Whether stating a correct or incorrect solution only a minority used the statistical term "independence" as the reason for their answer.



Question Number	Scheme	Marks
(a)	$Y \sim \text{Po}(0.25)$	B1
	$P(Y=0) = e^{-0.25}$ $= 0.7788$	M1 A1 (3)
(b)	$X \sim \text{Po}(0.4)$ $P(\text{Robot will break down}) = 1 - P(X=0)$ $= 1 - e^{-0.4}$ $= 1 - 0.067032$ $= 0.3297$	B1 M1 A1 (3)
(c)	$P(X=2) = \frac{e^{-0.4} (0.4)^2}{2!}$ $= 0.0536$	M1 A1 (2)
(d)	0.3297 or answer to part (b) as Poisson events are <u>independent</u>	B1ft B1 dep (2)
		Total [10]
	Notes (a) B1 for seeing or using $\text{Po}(0.25)$ M1 for finding $P(Y=0)$ either by e^{-a} , where a is positive (a needn't equal their λ) or using tables if their value of λ is in them Beware common Binomial error using, $p = 0.05$ gives 0.7738 but scores B0 M0 A0 A1 awrt 0.779 (b) B1 for stating or a clear use of $\text{Po}(0.4)$ in part (b) or (c) M1 for writing or finding $1 - P(X=0)$ A1 awrt 0.33 (c) M1 for finding $P(X=2)$ e.g. $\frac{e^{-\lambda} \lambda^2}{2!}$ with their value of λ in or if their λ is in the table for writing $P(X \leq 2) - P(X \leq 1)$ A1 awrt 0.0536 (d) 1 st B1 their answer to part(b) correct to 2 sf or awrt 0.33 2 nd B1 need the word independent. This is dependent on them gaining the first B1 SC Use of Binomial. Mark parts a and b as scheme. They could get (a) B0,M0,A0 (b) B0 M1 A0 In part c allow M1 for " $C_2(p)^2(1-p)^{n-2}$ " with "their n " and "their p ". They could get (c) M1,A0 DO NOT GIVE for $p(x \leq 2) - p(x \leq 1)$ In (d) they can get the first B1 only. They could get (d) B1B0	

Note: part (d) became part (e). Part (d) was added by MM, with marking guidance as follows:
(d)

B1: Recognise the need to find $P(X=0)$ for some unknown λ and for $P(X=0) > 0.9$

M1: Express this as an inequality using the formula and attempt to solve for λ using any method.

A1: Convert their λ back to a time frame, achieving a value in the range 2 hours – 2 hours 10 mins inclusive.

Handwritten work showing the derivation of the inequality $P(X=0) > 0.9$ and the conversion of λ to a time frame.

$$\begin{aligned}
 &X \sim \text{Po} - (1 / 20 \text{ hrs}) \quad \downarrow \div 20 \\
 &X \sim \text{Po} \left(\frac{1}{20} / \text{hr} \right) \quad \downarrow \times 20 \\
 &X \sim \text{Po} \left(\frac{1}{2} / \text{hrs} \right) \quad \downarrow \times t \\
 &\text{want} \\
 &P(X=0) > 0.9 \quad \Rightarrow \text{use the formula.} \\
 &e^{-\lambda} \lambda^0 > 0.9 \\
 &e^{-\lambda} > 0.9
 \end{aligned}$$

Trial + error of $e^{-\lambda}$ (table) :-

$$e^{-0.12} = 0.8869$$

$$e^{-0.11} = 0.8958$$

$$e^{-0.10} = 0.9048$$

$$e^{-0.105} = 0.9003$$

$\begin{array}{c} -\lambda \\ \downarrow \\ \begin{array}{ccc} -0.110 & -0.105 & -0.100 \end{array} \\ \begin{array}{ccc} < 0.9 & > 0.9 \end{array} \end{array}$

$\lambda = 0.1$
 $\frac{t}{20} = 0.1$
 $t = 0.1 \times 20$

OR

Using logs: $e^{-\lambda} > 0.9$

$$\ln(e^{-\lambda}) > \ln(0.9)$$

$$-\lambda < -0.1054 \quad \leftarrow -ve$$

$x-1$	$\lambda > 0.1054$	$x-1$
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$$\frac{t}{20} > 0.1054$$

$$t > 2.107 \text{ hrs.} \quad \underline{\underline{2 \text{ hrs } 6 \text{ mins}}}$$

3) The number of heaters, H , bought during one day from *Warmup* supermarket can be modelled by a Poisson distribution with mean 0.7

(a) Calculate $P(H \geq 2)$

(1)

The number of heaters, G , bought during one day from *Pumraw* supermarket can be modelled by a Poisson distribution with mean 3, where G and H are independent.

(b) Show that the probability that a total of fewer than 4 heaters are bought from these two supermarkets in a day is 0.494 to 3 decimal places.

(2)

(c) Calculate the probability that a total of fewer than 4 heaters are bought from these two supermarkets on at least 5 out of 6 randomly chosen days.

(3)

This question proved to be very accessible to all students. Parts (a), (b) and (c) were generally well answered. The most common error in part (b) was to not include the distribution used to obtain the final answer and in part (c) to use the Poisson distribution rather than the binomial distribution.

In part (d) the hypotheses were generally stated with the correct parameter although a minority of students tried to write them in words. The most common errors appeared in the calculation where a common error was to use the mean of 3.7 or to calculate the $P(X = 14)$. Pleasingly conclusions were generally given in context for this question.



Question	Scheme		Marks	AOs
(a)	$P(H \geq 2) = 0.1558$ awrt <u>0.156</u>		B1	1.1b
			(1)	
(b)	$H \sim \text{Po}(0.7) \quad G \sim \text{Po}(3)$			
	$Y = H + G \rightarrow Y \sim \text{Po}(3.7)$		M1	3.4
	$P(Y \leq 3) = 0.494^*$		A1cso*	1.1b
			(2)	
(c)	$K \sim B(6, 0.494)$		M1	3.3
	$P(K \geq 5) = 1 - P(K \leq 4)$		M1	1.1b
	$= 1 - 0.896\dots$			
	$= 0.1039\dots$ awrt <u>0.104</u>		A1	1.1b
			(3)	
(d)	$H_0: \lambda = "3.7" \quad H_1: \lambda > "3.7"$		B1ft	2.5
	$J \sim \text{Po}(7.4)$		B1ft	1.1b
	Method 1	Method 2		
	$P(J \geq 14) = 1 - P(J \leq 13)$ $= 1 - 0.9804\dots$	$P(J \geq 12) = 0.0735\dots$ $P(J \geq 13) = 0.0391\dots$	M1	1.1b
	$= 0.0195\dots$	CR $J \geq 13$	A1	1.1b
	0.0195 < 0.05 or $14 \geq 13$ or 14 is in the critical region or 14 is significant or Reject H_0 . There is evidence at the 5% level of significance that the number of heaters brought in total from the two supermarkets has increased.		A1	2.2b
			(5)	
(11 marks)				
Notes				
(a) B1: awrt 0.156				
(b) M1: For combining distributions and use of Po(3.7) A1*cso: $P(Y \leq 3) = 0.494$ we need to see $P(Y \leq 3)$ or $P(Y < 4)$ allow different letters.				
(c) M1: Setting up a new model $B(6, 0.494)$ may be implied by a correct answer or ${}^6C_n (0.494)^n (0.506)^{6-n}$ M1: Using $1 - P(K \leq 4)$ A1: awrt 0.104				
(d) B1: Both hypotheses correct using λ or μ . ft "3.7" from their 3.7 in part (b) and allow $2 \times$ "their 3.7" Ignore any words B1: Realising that $\text{Po}(2 \times \text{"their 3.7"})$ is to be used. This may be stated or used. M1: writing or using $1 - P(J \leq 13)$ or $1 - P(J < 14)$ or if finding a CR for writing $P(J \geq 12) = 0.0735\dots$ and $P(J \geq 13) = 0.0391\dots$ A1: awrt 0.0195 or CR $J \geq 13$ or $J > 12$ A1: A fully correct solution and drawing a correct inference in context.				

4) An engineering company manufactures an electronic component. At the end of the manufacturing process, each component is checked to see if it is faulty. Faulty components are detected at a rate of 1.5 per hour.	
(a) Suggest a suitable model for the number of faulty components detected per hour.	(1)
(b) Describe, in the context of this question, two assumptions you have made in part (a) for this model to be suitable.	(2)
(c) Find the probability of 2 faulty components being detected in a 1 hour period.	(2)
(d) Find the probability of at least one faulty component being detected in a 3 hour period.	(2)

This question was quite well done. Most candidates were aware of the conditions for a Poisson distribution but many lost marks as they did not answer the question in context, although the examination question actually specified this. A few did not realise that independent and random were the same condition. Cumulative probability tables were well used in part (c) and there were many accurate solutions using the Poisson formula.



Most candidates used a mean of 4.5 in part (d) and there were many accurate results.

Question Number	Scheme	Marks
(a)	$X \sim \text{Po}(1.5)$ need Po and 1.5	B1 (1)
(b)	Faulty components occur at a constant rate. Faulty components occur independently or randomly. Faulty components occur singly. any two of the 3 only need faulty once	B1 B1 (2)
(c)	$P(X=2) = P(X \leq 2) - P(X \leq 1)$ or $\frac{e^{-1.5}(1.5)^2}{2}$ $= 0.8088 - 0.5578$ $= 0.251$ awrt 0.251	M1 A1 (2)
(d)	$X \sim \text{Po}(4.5)$ 4.5 may be implied $P(X \geq 1) = 1 - P(X=0)$ $= 1 - e^{-4.5}$ $= 1 - 0.0111$ $= 0.9889$ awrt 0.989	B1 M1 A1 (3) (Total 8)

5) The number of cars passing an observation point in a 10 minute interval is modelled by a Poisson distribution with mean 1.

(b) Find the probability that in a randomly chosen 60 minute period there will be

- (i) exactly 4 cars passing the observation point,
- (ii) at least 5 cars passing the observation point.

(4)

The number of other vehicles, other than cars, passing the observation point in a 60 minute interval is modelled by a Poisson distribution with mean 12.

(c) Find the probability that exactly 1 vehicle, of any type, passes the observation point in a 10 minute period.

(3)

Most candidates were able to attempt part (b) successfully as these were fairly standard calculations. However, when required to apply Poisson probabilities to a problem it was only the better candidates who attained any marks in part (c).

(a) A sizeable proportion of candidates, whilst having learnt the conditions for a Poisson distribution, failed to realise that this applied to the events occurring. There were references to 'trials' and 'things' in some solutions offered.

b(i) Most candidates recognised Po(6) and were able to answer this successfully, either from the tables or by calculation. Common errors were using incorrect values from the table or calculating the exact value incorrectly.

(ii) Although many candidates attained full marks for this part, some were unable to express 'at least 5' correctly as an inequality and used $P(X \leq 5)$.

(b) Many of the successful candidates used a Po(3) to give a correct solution. Of the rest most candidates failed to realise that there were two ways for exactly one vehicle to pass the point and so only performed one calculation. This was often for P(1 car) and P(1 other vehicle), which were then added together. However there were some candidates who gained the correct answer via this method.



Question Number	Marks	Scheme	
(a)	Events occur at a constant rate. Events occur independently or randomly. Events occur singly.	any two of the 3	B1 B1 (2)
(b)	Let X be the random variable the number of cars passing the observation point.		
(i)	Po(6) $P(X \leq 4) - P(X \leq 3) = 0.2851 - 0.1512$ or $\frac{e^{-6} 6^4}{4!}$ $= 0.1339$		B1 M1 A1
(ii)	$1 - P(X \leq 4) = 1 - 0.2851$ or $1 - e^{-6} \left(\frac{6^4}{4!} + \frac{6^3}{3!} + \frac{6^2}{2!} + \frac{6}{1!} + 1 \right)$ $= 0.7149$		M1 A1 (5)
(c)	P (0 car and 1 others) + P (1 cars and 0 other) $= e^{-1} \times 2e^{-2} + 1e^{-1} \times e^{-2}$ $= 0.3679 \times 0.2707 + 0.3674 \times 0.1353$ $= 0.0996 + 0.0498$ $= 0.149$ alternative $P_o(1+2) = P_o(3)$ B1 $P(X=1) = 3e^{-3}$ M1 A1 $= 0.149$ A1		B1 M1 A1 A1 (4)
Notes			
(a)	B1 B1 Need the word events at least once. Independently and randomly are the same reason. Award the first B1 if they only gain 1 mark Special case. If they have 2 of the 3 lines without the word events they get B0 B1		
(b) (i)	B1 Using Po(6) in (i) or (ii) M1 Attempting to find $P(X \leq 4) - P(X \leq 3)$ or $\frac{e^{-\lambda} \lambda^4}{4!}$ A1 awrt 0.134		
(ii)	M1 Attempting to find $1 - P(X \leq 4)$ A1 awrt 0.715		
(c)	B1 Attempting to find both possibilities. May be implied by doing $e^{-\lambda_1} \times \lambda_2 e^{-\lambda_2} + e^{-\lambda_2} \times \lambda_1 e^{-\lambda_1}$ any values of λ_1 and λ_2 M1 finding one pair of form $e^{-\lambda_1} \times \lambda_2 e^{-\lambda_2}$ any values of λ_1 and λ_2 A1 one pair correct A1 awrt 0.149 Alternative. B1 for Po(3) M1 for attempting to find $P(X=1)$ with Po(3) A1 $3e^{-3}$ A1 awrt 0.149		